# 1. Vectors

## 1.1. Vector Notation

- A vector can be represented as $\overrightarrow{\mathrm{AB}}$ or $\mathbf{a}$
- The column vector form:

$$\overrightarrow{\mathrm{AB}} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

- The linear vector form:

$$\overrightarrow{\mathrm{AB}} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

## 1.2. Calculations with vectors

- Addition and Subtraction: Add or subtract each value of the vector with is corresponding value (**i** value with **i** value & **j** value with **j** value etc.)

$$(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) + (a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) = (x+a)\mathbf{i} + (y+b)\mathbf{j} + (z+c)\mathbf{k}$$

- Multiplication by a scalar: Multiply each value of the vector by the given value.

$$2\,(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \mathbf{2}x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$$

- Magnitude of a vector: Length of the vector

  Magnitude of $\overrightarrow{\mathrm{AB}}$ = $\left|\overrightarrow{\mathrm{AB}}\right| = \sqrt{x^2 + y^2 + z^2}$

- Unit vector: a vector that has a magnitude of 1

  Unit Vector of $\overrightarrow{\mathrm{AB}}$ = $\frac{1}{\left|\overrightarrow{\mathrm{AB}}\right|}\overrightarrow{\mathrm{AB}}$

- Displacement vector: Vector whose magnitude is the shortest distance between the two points. It is a straight line from one point to the other.
- Position vector: Position of a point relative to the origin. It is a straight line from the origin to a point. The position vector of point A is represented as $\overrightarrow{\mathrm{OA}}$.
- Dot product: Dot product of vectors $\mathbf{a}$ and $\mathbf{b}$ is written as $\mathbf{a}.\mathbf{b}$ ,and it can be calculated in two ways.

Method 1:

$$a.\mathrm{b} = xc + yd + ze$$

$$a.b = |a|\,|b| \cos\theta$$

Method 2: use the equation $a, b = |a||b| \cos\theta$

where cos$\theta$ = the angle between the two vectors

$|a|$ = magnitude of vector $\mathbf{a}$

$|b|$ = magnitude of vector $\mathbf{b}$

## 1.3. Equation of a Line

- The equation of a straight line is expressed in the form:

  r = a $+ t$ b

- For example:

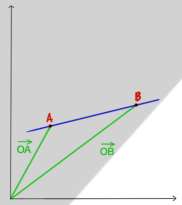
  The column vector form:

$$r = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

The linear vector form:

$$r = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k} + t\,(\mathbf{i} + \mathbf{j} + 3\mathbf{k})$$

## 1.4. Finding the Equation of a Line




- To find the equation of the line, given 2 points A and B:
  - Find the direction vector using

$$AB = OB - OA$$

- Substitute the values into the equation:

  r = a $+ t$ b

  where a = point A

  b = direction vector (vector AB)

  t = some scalar

## 1.5. Parallel, Skew or Intersects

For the two lines:

$\overrightarrow{\mathrm{OA}} = \tilde{\mathbf{a}} + s\tilde{\mathbf{c}} \quad \overrightarrow{\mathrm{OB}} = \tilde{\mathbf{b}} + t\tilde{\mathbf{d}}$

- Parallel:
  - For the lines to be parallel $\tilde{\mathbf{c}}$ must equal $\tilde{\mathbf{d}}$ or be in some ratio to it e.g. $1:2$
- Intersects:
  - Make $\overrightarrow{\mathrm{OA}} = \overrightarrow{\mathrm{OB}}$
  - If simultaneous works then intersects
  - If unknowns cancel then no intersection
- Skew:
  - First check whether line parallel or not
  - If not, then make $\overrightarrow{\mathrm{OA}} = \overrightarrow{\mathrm{OB}}$
  - Carry out simultaneous
  - When a pair does not produce same answers as another, then lines are skew



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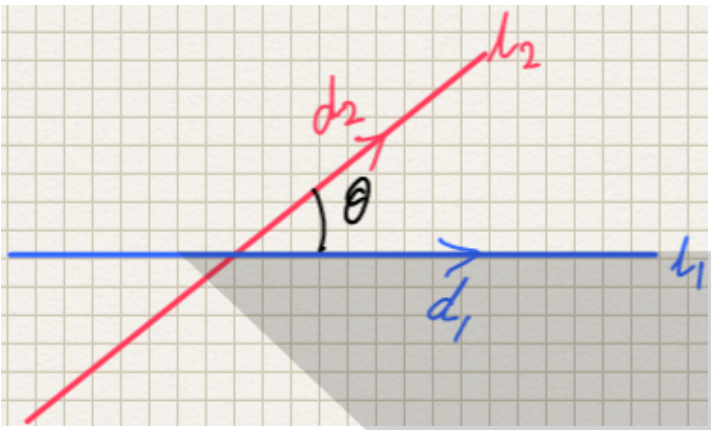
## 1.6. Angle between Two Lines

- Use dot product rule on the two direction vectors:

$$\frac{\mathrm{a.b}}{|a|\,|b|} = \cos\theta$$

- Note: $a$ and $b$ must be moving away from the point at which they intersect




## 1.7. ⊥ Distance from a Line to a Point

- **AKA:** shortest distance from a point to the line
- Find vector for the point, $\mathrm{B}$, on the line

Vector equation of the line:

$$\tilde{\mathbf{r}} = \begin{pmatrix}1\\3\\-2\end{pmatrix} + t\begin{pmatrix}1\\1\\3\end{pmatrix}$$

$$\therefore \overrightarrow{\mathrm{OB}} = \begin{pmatrix}1+t\\3+t\\3t-2\end{pmatrix}$$

- $A$ is the point given

$$\overrightarrow{\mathrm{OA}} = \begin{pmatrix}2\\3\\4\end{pmatrix}$$

$$\therefore \overrightarrow{\mathrm{AB}} = \begin{pmatrix}1+t-2\\3+t-3\\3t-2-4\end{pmatrix} = \begin{pmatrix}t-1\\t\\3t-6\end{pmatrix}$$

- Use Dot product of $\mathrm{AB}$ and the direction vector

$$\overrightarrow{\mathrm{AB}}.\mathbf{d} = \cos 90$$

$$\begin{pmatrix}t-1\\t\\3t-6\end{pmatrix}.\begin{pmatrix}1\\1\\3\end{pmatrix} = 0$$

$$1\,(t-1) + 1\,(t) + 3\,(3t-6) = 0$$

$$11t - 19 = 0$$

$$t = \frac{19}{11}$$

- Substitute $t$ into equation to get foot
- Use Pythagoras' Theorem to find distance

**{S08-P3} Question:**

The points $A$ and $B$ have position vectors, relative to the origin $O$, given by

$OA = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $OB = 2\mathbf{i} + \mathbf{j} + 3\mathbf{k}$

The line $l$ has vector equation

$$\mathbf{r} = (1-2t)\,\mathbf{i} + (5+t)\,\mathbf{j} + (2-t)\,\mathbf{k}$$

i) Show that $l$ does not intersect the line passing through $A$ and $B$.

ii) The point $\mathrm{P}$ lies on $l$ and is such that angle $\mathrm{PAB}$ is equal to $60^\circ$. Given that the position vector of $P$ is $(1-2t)\,\mathbf{i} + (5+t)\,\mathbf{j} + (2-t)\,\mathbf{k}$, show that $3t^2 + 7t + 2 = 0$. Hence find the only possible position vector of $P$

**Solution:**

**Part (i)**

*Firstly, we must find the equation of line* $\mathrm{AB}$

$$AB = OB - OA$$

$$= \begin{pmatrix}2\\1\\3\end{pmatrix} - \begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}1\\-1\\0\end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix}1\\2\\3\end{pmatrix} + s\begin{pmatrix}1\\-1\\0\end{pmatrix}$$

and

$$\mathbf{L} = \begin{pmatrix}1\\5\\2\end{pmatrix} + t\begin{pmatrix}-2\\1\\-1\end{pmatrix}$$

*Equating the two lines*

$$\begin{pmatrix}1+s\\2-s\\3\end{pmatrix} = \begin{pmatrix}1-2t\\5+t\\2-t\end{pmatrix}$$

*Equation 1:* $1 + s = 1 - 2t$ so $s = -2t$

*Equation 2:* $2 - s = 5 + t$

*Substitute 1 into 2:*

$$2 + 2t = 5 + t$$

$\therefore t = 3$ and then $s = -6$

*Equation 3:*

$$3 = 2 - t$$

*Substitute the value of* $t$

$3 = 2 - 3$ so $3 = -1$

*This is incorrect therefore lines don't intersect*

**Part (ii)**



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*Angle* $\mathrm{PAB}$ *is formed by the intersection of the lines* $\mathrm{AP}$ *and* $\mathrm{AB}$

$$P = \begin{pmatrix} 1-2t \\ 5+t \\ 2-t \end{pmatrix}$$

$$AP = OP - OA$$

$$AP = \begin{pmatrix} 1-2t \\ 5+t \\ 2-t \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2t \\ 3+t \\ -1-t \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

*Now use the dot product rule to form an eqn.*

$$\frac{|\mathrm{AP.AB}|}{|\mathrm{AP}|\,|\mathrm{AB}|};\ \frac{-3t-3}{\sqrt{6t^2+8t+10}\times\sqrt{2}} = \cos 60$$

$$3t-3 = \frac{1}{2}\sqrt{6t^2+8t+10}\times\sqrt{2}$$

$$36t^2+72t+36 = 12t^2+16t+20$$

$$24t^2+56t+16 = 0$$

$t = -\frac{1}{3}$ or $t = -2$

**{W11-P31} Question:**

With respect to the origin $O$, the position vectors of two points $A$ and $B$ are given by $\overrightarrow{\mathrm{OA}} = \mathbf{i}+2\mathbf{j}+2\mathbf{k}$ and $\overrightarrow{\mathrm{OB}} = 3\mathbf{i}+4\mathbf{j}$ . The point $P$ lies on the line through $A$ and $B$, and $\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{AB}}$

i) $\overrightarrow{\mathrm{OP}} = (1+2\lambda)\,\mathbf{i}+(2+2\lambda)\,\mathbf{j}+(2-2\lambda)\,\mathbf{k}$

ii) By equating expressions for $\cos\mathrm{AOP}$ and $\cos\mathrm{BOP}$ in terms of $\lambda$, find the value of $\lambda$ for which $\mathrm{OP}$ bisects the angle $\mathrm{AOB}$ .

**Solution:**

**Part (i)**

$$\overrightarrow{\mathrm{AP}} = \lambda\overrightarrow{\mathrm{AB}} = \lambda\,(OB-OA)$$

$$= \lambda\begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}$$

$$\therefore AP = \begin{pmatrix} 2\lambda \\ 2\lambda \\ -2\lambda \end{pmatrix}$$

$$OP = OA + \begin{pmatrix} 2\lambda \\ 2\lambda \\ -2\lambda \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 2\lambda \\ 2\lambda \\ -2\lambda \end{pmatrix}$$

**Part (ii)**

*Interpreting the question gives the information that* $\mathrm{AOP}$ *is equal to* $BOP \therefore$ $\cos\mathrm{AOP}$ *is equal to* $\cos\mathrm{BOP}$ *. Now you can equate the two dot product equations*

$$\cos\mathrm{AOP} = \frac{\mathrm{OA.OP}}{|\mathrm{OA}|\,|\mathrm{OP}|} = \frac{9+2\lambda}{3\sqrt{9+4\lambda+12\lambda^2}}$$

$$\cos\mathrm{BOP} = \frac{\mathrm{OB.OP}}{|\mathrm{OB}|\,|\mathrm{OP}|} = \frac{11+14\lambda}{5\sqrt{9+4\lambda+12\lambda^2}}$$

$$\frac{9+2\lambda}{3\sqrt{9+4\lambda+12\lambda^2}} = \frac{11+14\lambda}{5\sqrt{9+4\lambda+12\lambda^2}}$$

*Cancel out the denominator to give you*

$$\frac{9+2\lambda}{3} = \frac{11+14\lambda}{5}$$

$$45+10\lambda = 33+42\lambda$$

$12 = 32\lambda$ and $\therefore \lambda = \frac{3}{8}$



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